Causal Inference

1.

a) Post hoc fallacy.

b) It is merely a coincidence. The professor's forgetting is likely due to other factors, not your presence.

2.

a) No.

b) For each individual unit, we can observe only one of the two counterfactual outcomes.

3.

a) Let

$$\boldsymbol{A} = g_{\boldsymbol{A}}(\boldsymbol{\varepsilon_A}), \qquad \boldsymbol{S} = g_{\boldsymbol{S}}(\boldsymbol{A}, \boldsymbol{Y}, \boldsymbol{\varepsilon_S}), \qquad \boldsymbol{Y} = g_{\boldsymbol{Y}}(\boldsymbol{\varepsilon_Y})$$

where $g_{\boldsymbol{A}}(\cdot), g_{\boldsymbol{Y}}(\cdot)$ and $g_{\boldsymbol{S}}(\cdot)$ are unknown, non-parametric functions, and $\boldsymbol{\varepsilon_A}, \boldsymbol{\varepsilon_Y}$ and $\boldsymbol{\varepsilon_S}$ are mutually independent exogenous error terms.

b) Simply consider

$$\boldsymbol{A} \sim U(-2{,}2), \qquad \boldsymbol{Y} \sim U(-2{,}2)$$

and define

$$g_{\boldsymbol{S}}(\boldsymbol{A}, \boldsymbol{Y}, \boldsymbol{\varepsilon_S}) = \boldsymbol{Y} + \boldsymbol{A} + \boldsymbol{\varepsilon_S}$$

When we control for $\boldsymbol{S} = s$ then $s = \boldsymbol{Y} + \boldsymbol{A} + \boldsymbol{\varepsilon_S}$. Now, the value of $\boldsymbol{Y}$ depends on the value of $\boldsymbol{A}$**.** Therefore, conditioning on the $\boldsymbol{S}$ induces a statistical association between $\boldsymbol{A}$ and $\boldsymbol{Y}$, and the statement $\boldsymbol{A} \perp \boldsymbol{Y} | \boldsymbol{S}$ is false.

4.

a) By pair definition,

$$\boldsymbol{Y}_i = \boldsymbol{Y}_i(0)(1 - \boldsymbol{A}_i) + \boldsymbol{Y}_i(1)\boldsymbol{A}_i$$

then

i	$\boldsymbol{A}_i$	$\boldsymbol{Y}_i$	$\boldsymbol{Y}_i(1)$	$\boldsymbol{Y}_i(0)$	Type
1	1	1	1	1	Always − pass
2	1	0	0	0	Always − fail
3	0	0	1	0	Helped by tutoring
4	0	1	0	1	Harmed by tutoring
5	1	1	1	1	Always − pass

b) For Causal Effect (Average Treatment Effect)

$$E[\boldsymbol{Y}(1)] - E[\boldsymbol{Y}(0)] = \frac{3}{5} - \frac{3}{5} = 0$$

For Association (Observed Difference in Mean)

$$E(\boldsymbol{Y}|\boldsymbol{A} = 1) - E(\boldsymbol{Y}|\boldsymbol{A} = 0) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

The core reason for this difference is selection bias: The fact that the assignment mechanism is not random is the source of the selection bias. $E[\boldsymbol{Y}(a)]$ cannot identified as $E(\boldsymbol{Y}|\boldsymbol{A} = a)$.

5.

a) Exogeneity Condition

$$E(\boldsymbol{X}^{\top}\boldsymbol{\varepsilon}) = \boldsymbol{0}$$

b) Since the OLS estimator of $\boldsymbol{\beta}$ is

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y}$$

then the expectation of $\widehat{\boldsymbol{\beta}}$ is unbiased if exogeneity condition hold as follow

$$E(\widehat{\boldsymbol{\beta}}) = E[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y}] = E[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top (\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})]$$
$$= \boldsymbol{\beta} + (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{E}(\boldsymbol{X}^\top \boldsymbol{\varepsilon}) = \boldsymbol{\beta}$$

6.

a) Consider a set of nodes $\boldsymbol{C} = (v_1, \cdots, v_N)$ such that $\boldsymbol{C} \cap \boldsymbol{A} = \emptyset$. Assume that following conditions hold:

i) No elemet of $\boldsymbol{C}$ is a descendent of $\boldsymbol{A}$

ii) $\boldsymbol{C}$ blocks all backdoor paths from $\boldsymbol{A}$ to $\boldsymbol{Y}$

Then $\boldsymbol{Y}(a) \perp \boldsymbol{A}|\boldsymbol{C}\ \forall a$

b) There is an arrow $\boldsymbol{C} \longrightarrow \boldsymbol{A}$. The presence of this arrow creates an open backdoor path $\boldsymbol{A} \longleftarrow \boldsymbol{C} \longleftarrow \boldsymbol{U}_1 \longleftarrow \boldsymbol{Y}$ because $\boldsymbol{U}_1$ is a common cause of $\boldsymbol{A}$ and $\boldsymbol{Y}$, and so confounding exists. Conditioning on $\boldsymbol{C}$ would block the backdoor path but simultaneously open a backdoor path on which C is collider ($\boldsymbol{A} \longleftarrow \boldsymbol{U}_2 \longrightarrow \boldsymbol{C} \longleftarrow \boldsymbol{U}_1 \longrightarrow \boldsymbol{Y}$)

7.

a) The partial correlation

$$\rho_{\boldsymbol{YA}|\boldsymbol{C}} = \frac{\rho_{\boldsymbol{YA}} - \rho_{\boldsymbol{YC}}\rho_{\boldsymbol{CA}}}{\sqrt{(1-\rho_{\boldsymbol{YC}}^2)(1-\rho_{\boldsymbol{CA}}^2)}} = \frac{\frac{17}{25} - \frac{4}{5} \times \frac{3}{5}}{\sqrt{\left(1-\left[\frac{4}{5}\right]^2\right)\left(1-\left[\frac{3}{5}\right]^2\right)}} = \frac{5}{12}$$

b) It is easy to see that

$$\rho_{\boldsymbol{YA}|\boldsymbol{C}} = \frac{5}{12} < \frac{17}{25} = \rho_{\boldsymbol{YA}}$$

The link between study hours and exam scores weakened from $17/25$ to $5/12$ after controlling the effect of stress. This means stress was a confounder that made the original relationship look stronger than it truly was.

8.

a) The bookmaker's margin is

$$\left(\frac{2}{3} + \frac{5}{11}\right) - 1 = \left(\frac{22+15}{33}\right) - 1 = \frac{4}{33}$$

b) The fair odds are

$$\text{Buakaw to win:} \left(\frac{\frac{2}{3}}{\frac{37}{33}}\right)^{-1} = \left(\frac{22}{37}\right)^{-1} = \frac{37}{22}$$

$$\text{Fighter X to win:} \left(\frac{\frac{5}{11}}{\frac{37}{33}}\right)^{-1} = \left(\frac{15}{37}\right)^{-1} = \frac{37}{15}$$